THE DUFFING OSCILLATOR

Dr. Florian Bociort
f.bociort@tudelft.nl, Room 22E020

1. INTRODUCTION

The motion of a particle in one dimension subject to a force F, depending on the particle's position x and velocity $\dot{x}$ and on time t, is determined by Newton's equation of motion:

$$m\ddot{x}(t) = F\left[x(t), \dot{x}(t), t\right]. \tag{1.1}$$

The (double) dot denotes a (double) derivative with respect to time. A solution can be found for each set of initial conditions $x(t_0)$ and $\dot{x}(t_0)$ given at some time t_0. Analytical solutions exist for constant force, for the harmonic oscillator ($F = -\kappa x$), and for a number of other cases. In Section 2 a simple numerical method for solving this equation is described and this can be applied straightforwardly to arbitrary forces and initial conditions.

Interesting and sometimes surprising physical phenomena can now be studied. As an example, consider the Duffing oscillator, with a force given by

$$F\left[x, \dot{x}, t\right] = -\gamma\dot{x} \,+\, 2ax \,-\, 4bx^3 + F_0\cos(\omega t). \tag{1.2}$$

The first term on the right hand side represents a velocity-dependent friction; the second and third terms are the force a particle feels when it moves in a double potential well $bx^4 - ax^2$, and the last term is an external periodic force. An experimental realisation is a pendulum consisting of an iron ball suspended by a thin string, with two magnets below it. The pendulum and the magnets are placed on a table which is moved back and forth with frequency ω. The string and the air provide the frictional force, the two magnets together with gravity form some kind of double potential well, and, in the reference frame in which the pendulum is at rest, the periodic motion of the table is felt as a periodic force. It turns out that the Duffing oscillator exhibits *chaotic behavior* for particular values of the parameters γ, a, b, F_0 and ω.

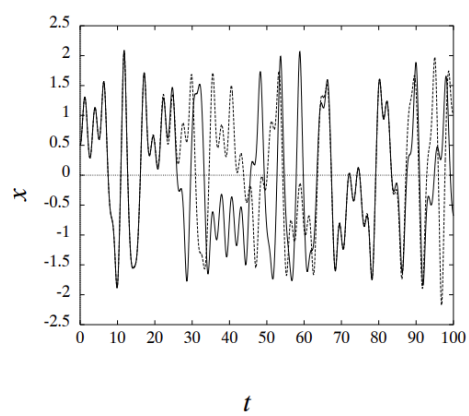

Figure 1. Solution of the Duffing oscillator. Parameters are $m = 1$, $a = 1/2$, $b = 1/4$, $F_0 = 2.0$, $\omega = 2.4$, $\gamma = 0.1$. Two solutions are shown: one with initial position $x_0 = 0.5$, the other with $x_0 = 0.5001$ ($\dot{x}_0 = 0$ in both cases). For these nearly equal initial conditions, the solutions soon become uncorrelated, showing the difficulty in predicting the time evolution of a chaotic system.

This means that the motion itself looks irregular and that a very small change in the initial conditions will grow and result in a completely different motion. Figure 1 shows the behavior of the Duffing oscillator for two nearly equal initial conditions, showing the sensitivity to these conditions.

Over the last few decades chaotic systems have been studied extensively. A system which often behaves chaotically is the weather. The difficulty in predicting the evolution of chaotic systems causes weather forecasts to be increasingly unreliable as they look further into the future, and occasionally to be drastically wrong.

2. VERLET ALGORITHM

The Verlet algorithm is a simple method for integrating second-order differential equations of the form

$$\ddot{x}(t) = F[x(t), t]. \tag{2.1}$$

It had been mentioned already at the end of the eighteenth century and is used very often for classical mechanics problems because of its simplicity and stability. It has a fixed time discretisation interval and needs only one evaluation of the force F per step. The Verlet algorithm is easily derived by adding the Taylor expansions for the coordinate x at $t = \pm h$ about $t = 0$:

$$x(h) = x(0) + h\dot{x}(0) + \frac{h^2}{2} F\left[x(0), 0\right] + \frac{h^3}{6} \dddot{x}(0) + \mathrm{O}\left(h^4\right) \tag{2.2}$$

$$x(-h) = x(0) - h\dot{x}(0) + \frac{h^2}{2} F\left[x(0), 0\right] - \frac{h^3}{6} \dddot{x}(0) + \mathrm{O}\left(h^4\right) \tag{2.3}$$

leading to

$$x(h) = 2x(0) - x(-h) + h^2 F\left[x(0), 0\right] + \mathrm{O}\left(h^4\right). \tag{2.4}$$

Knowing the values of x at time 0 and $-h$, this algorithm predicts the value of $x(h)$. Here we need the last two values of x to produce the next one. But usually we have the initial position, $x(0)$, and velocity, $v(0)$, at our disposal, so we change the procedure. We approximate $x(h)$ by

$$x(h) = x(0) + v(0)h + \frac{h^2}{2} F\left[x(0), 0\right] + \mathrm{O}\left(h^3\right). \tag{2.5}$$

The algorithm is invariant under time reversal, as is the original differential equation.

The accumulated error in the position after a large number of integration steps is of order $\mathrm{O}\left(h^2\right)$. As there is no point in calculating the velocities with higher precision than that of the positions, we can evaluate the velocities using the simple formula

$$v(0) = \frac{x(h) - x(-h)}{2h} \tag{2.6}$$

which gives the velocity with the required $\mathrm{O}\left(h^2\right)$ error.

Using the velocities at half integer time steps:

$$v[(n + 1/2)h] = \{x[(n + 1)h] - x(nh)\}/h + \mathrm{O}\left(h^2\right), \tag{2.7}$$

we may reformulate the Verlet algorithm in the so-called *"leap-frog"* form:

$$v(h/2) = v(-h/2) + hF\left[x(0), 0\right] \tag{2.8}$$

$$x(h) = x(0) + hv(h/2). \tag{2.9}$$

Verify that this form is exactly equivalent to the Verlet form in Eq. (2.4). The error in the positions is therefore still $\mathrm{O}\left(h^4\right)$.

However, to start from $x(0)$ and $v(0)$ we need not only half-step velocity evaluations as in Eq. (2.8) but also full-step velocities. The time step is not easily adapted in the Verlet/leap-frog algorithm. In the most convenient "velocity-Verlet" form, we use different ways to evaluate F in these two cases.

To compute the half-step velocities we write:

$$v(t+h/2) = v(t) + h/2F(t); \tag{2.10}$$
$$x(t+h) = x(t) + hv(t+h/2); \tag{2.11}$$

For full-step velocities we write:

$$v(t+h) = v(t+h/2) + h/2F(t+h). \tag{2.12}$$

Consider the Verlet solution for the harmonic oscillator $\ddot{x} = -C\,x$:

$$x(t+h) = 2x(t) - x(t-h) - h^2 C\,x(t) \tag{2.13}$$

to see where numerical care is needed. The analytic solution to this algorithm (not the exact solution to the continuum differential equation) can be written in the form $x(t) = \exp(i\,\omega t)$, with ω satisfying

$$2 - 2\cos(\omega h) = h^2 C. \tag{2.14}$$

Verify that if $h^2 C > 4$, ω becomes imaginary and the analytical solution becomes unstable. Of course we would not take h that large in actual applications, as we always take it substantially smaller than the period of the continuum solution. However, this instability also occurs when integrating systems of second order differential equations where the solution has high-frequency Fourier components. The step size must then be sufficiently small.

For the Duffing oscillator, the force $F[x,\, v,\, t]$ in Eq. (1.2) also includes the velocity. We therefore adapt the Verlet algorithm: we separate the friction term $-\gamma v$ from the rest of F and include it in the full-step velocity terms, using again

$$v(t) = \frac{x(t+h) - x(t-h)}{2h}. \tag{2.15}$$

This leads to an adapted form of the algorithm:

$$v(t+h/2) = (1-\gamma h/2)v(t) + h/2F\left(x(t), t\right); \tag{2.16}$$
$$x(t+h) = x(t) + hv(t+h/2); \tag{2.17}$$
$$\text{Calculate } F(x(t+h), t+h); \tag{2.18}$$
$$v(t+h) = \frac{v(t+h/2) + h/2\,F\left(x(t+h), t+h\right)}{1+\gamma h/2} \tag{2.19}$$

Verify this. $F(x,\, t)$ is here the result after removing the velocity term from $F[x,\, v,\, t]$ in Eq. (1.2). Use the mass $m = 1$.

3. YOUR TASK

Write a short report (about 5-6 A4 pages) containing:
- an introduction: search on the Internet for chaos, nonlinear systems, etc.,
- the algorithm for solving Eq. (1.2)
- the 3 verifications required in Section 2 (marked as Verify), and
- numerical results for the Duffing oscillator, as follows:

a) Implement the Verlet algorithm (Eqs. (2.16)-(2.19)) in a computer program in Python (or in another programming language if you wish). First, show numerically that for a conservative Hamiltonian, this algorithm preserves the energy. For this, remove the terms from the Hamiltonian which are non-conservative. Which parameters in Eq. (1.2) must then be zero? Does the numerical error of the energy computation increase in time or not?

b) Solve Eq. (1.2) numerically for $F_0 = 2.0$, $\omega = 2.4$, $\gamma = 0.1$ and initial conditions in the vicinity of $x_0 = 0.5$, $\dot{x}_0 = 0$. Plot a few curves in the $(t,\, x)$ and $(x,\, p)$ plane, showing the extreme sensitivity of the results with small changes in initial conditions. (Keep $a = 1/2$, $b = 1/4$ but you can also try different values of the other parameters.)

c) For the parameter values given at b), plot the values of x and $p = \dot{x}$ each time a period $T = 2\pi/\omega$ has elapsed in a x/p diagram. Plot many points! The points you will get form a *strange attractor*. Zoom in to this set of points and describe what you see.

It is possible to assign a dimension to such a structure. This is done by covering the set of points with grids of different sizes, and counting the number of squares $N(b)$ needed to cover the structure with squares of size $b \times b$. For sufficiently small values of b this number then depends on b as

$$N\,(b) \sim b^{-D_f}\,,$$

and the exponent D_f is called the dimension. When this number is not an integer, we call the dimension, and the corresponding structure, *fractal*.

d) Argue that the dimension of a line segment determined in this way is equal to one, and that that of a square surface is equal to two.

e) Determine the fractal dimension of the strange attractor constructed at c). This can be done as follows (other programming approaches are also possible):

1) Start with the array(s) containing all points of the attractor that you have used for the plot at c). First, normalize the x and v values of the attractor points to [0,1]. The attractor then lies in a square of side 1.

2) Divide this square into smaller $l \times l$ square cells, where l is first taken to be 2, then 4, and so on up to $l = 128$. The side of the cells is then $b = 1/l$. A new $l \times l$ array assumes the value 1 for cells which contain a point of the attractor and 0 if that is not the case. (Programming hint: To fill the array easily, transform the attractor array(s) for each l value so that all attractor points get a position described by two positive integers $< l$). When this array is filled, $N(b)$ is the number of array elements equal to 1.

3) Plot the resulting $\log[N(b)]$ versus $\log(b)$. For an attractor of 25000 points, these results lie more or less on a straight line with a slope $-D_f$ which then gives the fractal dimension. Is the dimension you have obtained an integer or not?

- Together with your report, email your computer code to the instructor for plagiarism check (if necessary in a .zip file, but not a link to a file somewhere else).

4. JUST FOR FUN

Chaos and fractals can produce nice images, see e.g. `https://home.imphys.tudelft.nl/~fbociort/fractals.html`.

The colorful images shown there come from the design of optical imaging systems. The instructor of this assignment and his former PhD student have discovered in the most widespread software implementation of optical system optimization a complicated form of chaos (a so-called non-attracting chaotic saddle). The presence of such fractals (and chaotic saddles) makes optical design more difficult. This discovery was a surprise for experts and has lead to algorithmic improvements in state-of-the-art commercial optical design software.

5. START

Read first this entire document carefully. Section 3 tells you what you have to do. If you have specific questions about the requirements listed there, send me an email to schedule a meeting (physical or online).

Otherwise, after you inform me that you have registered you can start with this theoretical assignment immediately. We meet only if you need help. If you need help with coding, make an appointment with my PhD student Kumar Rishav (email: K.Rishav@tudelft.nl). Your degree of independence will be reflected in your mark.

It is important that you send me your first version of your report according to the RP administration deadline for your round. Do not forget to send me your report via email! (Submission via Brightspace is not sufficient!)

Enjoy the Research Practicum!